1. Explain the difference between a correlated subquery and a nested subquery.

- **Answer:** A correlated subquery references data from the outer query in its WHERE clause. A nested subquery can be placed anywhere in the outer query and doesn't directly reference the outer table.

3. How can you optimize a slow-running query?

- **Answer:** There are several techniques, including:
    - Using appropriate indexes
    - Avoiding unnecessary joins and subqueries
    - Using efficient functions and operators
    - Analyzing execution plans to identify bottlenecks

4. Explain the concept of normalization in database design.

- **Answer:** Normalization is the process of organizing tables to minimize redundancy and improve data integrity. It involves breaking down tables into smaller, more focused tables with defined relationships.

5. Describe the purpose of window functions in SQL.

- **Answer:** Window functions perform calculations on a set of rows within a query partition. Examples include ranking, calculating moving averages, and finding percentiles.

7. How can you handle missing values (NULL) in your queries?

- **Answer:** You can use functions like ISNULL, COALESCE, or CASE statements to handle missing values. These functions allow you to specify alternative values or perform different operations based on NULL checks.

8. Explain the concept of materialized views and their benefits.

- **Answer:** Materialized views are pre-computed copies of database queries stored as tables. They improve query performance by providing pre-aggregated or filtered data, reducing processing needs for repeated queries.

9. Describe the concept of transactions in SQL and their ACID properties.

- **Answer:** A transaction is a unit of work that maintains data consistency. ACID stands for Atomicity (all or nothing), Consistency (data integrity), Isolation (concurrent transactions don't interfere), and Durability (changes persist).

10. How can you secure your SQL queries against SQL injection attacks?

- **Answer:** Use prepared statements with parameterized queries. These statements separate code from data, preventing malicious code execution.

11. Explain the difference between UNION and UNION ALL in SQL.

- **Answer:** UNION removes duplicate rows from the combined result set, while UNION ALL includes all rows, even duplicates, from the selected queries.

12. Describe the concept of common table expressions (CTEs) and their benefits.

- **Answer:** CTEs are temporary named result sets defined within a query. They improve readability and modularity by allowing complex logic to be pre-defined and reused within the main query.

13. Explain the concept of temporal tables and how they can be used for historical data tracking.

- **Answer:** Temporal tables store data with historical versions, allowing you to track changes over time. You can use them to analyze historical trends, audit data modifications, or revert to previous states.

14. Describe the functionality of triggers in SQL and their different types.

- **Answer:** Triggers are stored procedures that automatically execute in response to specific events on a table (like INSERT, UPDATE, DELETE). They can be used for data validation, enforcing business logic, or maintaining data consistency.

15. Explain the difference between full outer joins and full joins in SQL.

- **Answer:** Both join types return all rows from both tables, but a full outer join preserves NULL values in unmatched columns, while a full join may replace them with default values depending on the database engine.

16. Describe the purpose of regular expressions in SQL and how they can be used in queries.

- **Answer:** Regular expressions are patterns used to search and manipulate text data within SQL queries. They allow for complex string matching and manipulation, enabling tasks like data validation or extracting specific information.

17. Explain the concept of database partitioning and its benefits.

- **Answer:** Database partitioning divides a large table into smaller, manageable segments based on a chosen key. This improves performance by allowing queries to target specific partitions and reducing I/O operations.

18. Describe the functionality of MERGE statements in SQL.

- **Answer:** MERGE statements combine INSERT, UPDATE, and DELETE operations into a single statement. They offer efficient data manipulation, allowing you to perform conditional actions based on existence or matching criteria.